

LONDON LUTON, United Kingdom

WMO: 036733

Lat: 51.875N

Lon: 0.368W

Elev: 160

StdP: 99.41

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.8	(c) -2.2	(d) -5.2	(e) 2.5	(f) -1.4	(g) -4.1	(h) 2.7	(i) -0.7	(j) 14.7	(k) 8.1	(l) 13.2	(m) 8.1	(n) 3.8	(o) 30	(p) 0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.1	(c) 27.1	(d) 18.9	(e) 25.0	(f) 18.3	(g) 23.0	(h) 17.3	(i) 20.1	(j) 25.1	(k) 19.1	(l) 23.6	(m) 18.0	(n) 21.8	(o) 3.8	(p) 120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	13.4	22.4	17.2	12.5	20.7	16.2	11.8	19.5	58.8	25.5	55.2	23.4	51.8	21.5	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.7	10.2	9.1	-5.5	30.5	1.4	2.2	-6.5	32.1	-7.3	33.3	-8.1	34.5	-9.1	36.1
			-5.6	22.1	1.4	1.5	-6.6	23.2	-7.4	24.1	-8.2	24.9	-9.2	26.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	4.1	4.5	6.3	8.7	11.7	14.8	17.2	17.0	14.4	11.0	7.1	4.7
	DBStd	5.55	3.20	3.21	3.03	3.07	2.96	2.73	2.72	2.62	2.46	2.84	3.07	3.55
	HDD10.0	813	182	154	118	59	15	1	0	0	1	22	96	166
	HDD18.3	3027	440	387	373	289	205	110	55	58	122	228	338	422
	CDD10.0	875	1	1	4	20	69	146	224	217	131	52	8	2
	CDD18.3	47	0	0	0	0	1	6	21	17	3	0	0	0
	CDH23.3	386	0	0	0	1	5	57	176	129	17	2	0	0
	CDH26.7	81	0	0	0	0	0	9	40	31	1	0	0	0
Wind	WSAvg	4.6	5.3	5.3	5.1	4.5	4.4	4.2	4.0	4.0	4.2	4.5	4.6	5.1
Precipitation	PrecAvg	693	67	47	53	54	50	60	46	54	60	67	67	69
	PrecMax	888	137	109	129	118	124	131	104	120	137	199	156	157
	PrecMin	516	13	8	3	6	2	6	7	2	10	5	22	16
	PrecStd	100	32	28	29	28	28	35	23	28	38	43	32	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	13.2	16.2	21.8	24.0	28.0	30.2	29.9	25.2	20.2	15.2	13.2
		MCWB	11.1	10.6	11.2	14.4	16.8	19.3	19.8	20.3	18.7	16.7	13.8	11.9
	2%	DB	11.1	11.1	14.0	18.1	21.8	24.9	27.2	26.9	22.2	17.8	14.0	12.1
		MCWB	10.0	9.5	10.4	12.8	16.1	18.1	18.7	19.5	17.2	15.1	12.6	11.0
	5%	DB	10.0	10.1	12.2	15.9	19.8	22.2	25.1	24.1	20.2	16.1	12.9	11.1
		MCWB	9.1	8.9	9.6	11.6	14.8	16.7	18.1	18.2	16.0	14.3	11.7	10.0
	10%	DB	9.0	9.1	11.0	14.0	17.2	20.2	23.0	22.1	18.9	15.1	11.8	10.0
		MCWB	8.3	8.2	8.8	10.4	13.2	15.6	17.3	17.2	15.1	13.5	10.7	9.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	11.3	12.2	15.0	18.5	20.6	21.1	22.5	19.9	17.0	14.3	12.5
		MCDB	12.2	12.9	14.7	20.2	22.0	26.1	26.9	28.1	23.5	19.7	15.1	13.2
	2%	WB	10.4	10.1	11.2	13.3	16.5	18.9	20.0	20.2	18.2	15.7	13.0	11.6
		MCDB	11.1	11.0	13.4	17.3	20.4	23.5	25.5	24.8	21.4	17.3	13.8	12.3
	5%	WB	9.2	9.1	10.3	11.9	15.1	17.4	18.8	18.9	16.9	14.9	11.8	10.5
		MCDB	9.7	9.8	11.9	15.1	18.6	21.1	23.8	22.9	19.5	16.1	12.8	11.2
	10%	WB	8.2	8.2	9.3	10.7	13.9	16.2	17.7	17.7	15.7	13.9	10.9	9.3
		MCDB	8.8	9.0	10.7	13.4	16.8	19.6	21.8	21.0	17.9	15.0	11.8	9.9
Mean Daily Temperature Range	5% DB	MDBR	4.7	5.3	6.8	8.2	8.6	8.9	9.1	8.7	7.8	6.4	5.1	4.5
		MCDBR	5.7	6.0	8.6	11.2	12.0	12.3	12.9	12.0	10.0	7.1	5.3	5.1
		MCWBR	5.6	4.9	5.9	6.8	7.2	6.8	5.9	5.9	5.6	5.0	4.9	5.0
	5% WB	MCDBR	5.7	5.7	7.9	10.0	10.6	11.1	11.5	10.4	9.0	6.5	5.2	5.0
		MCWBR	5.7	5.1	6.0	6.5	6.9	6.9	6.3	6.1	5.8	5.2	5.1	5.1
Clear-Sky Solar Irradiance	taub	0.321	0.337	0.371	0.393	0.398	0.399	0.396	0.386	0.368	0.351	0.325	0.311	
	taud	2.378	2.391	2.289	2.242	2.264	2.285	2.301	2.351	2.404	2.438	2.424	2.359	
	Ebn at Noon	681	780	815	841	855	855	851	840	814	753	672	624	
	Edh at Noon	61	79	105	124	127	126	122	110	93	75	58	53	
All-Sky Solar Radiation	RadAvg	0.79	1.46	2.54	3.92	4.77	5.22	4.97	4.12	3.10	1.80	0.95	0.63	
	RadStd	0.07	0.19	0.33	0.44	0.31	0.51	0.51	0.32	0.27	0.16	0.11	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (84 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page